The smooth 23×80 model

A continuous, exactly Bayesian model of $d\sigma/dm_{hh}$ as the primary surrogate

Plan, for review before any code is written

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Abstract

The free 23×80 fit, 23 numbers per m_{hh} bin fitted by weighted least squares, is as accurate as the network ensemble, its posterior is exactly Gaussian, and it needs no training. Its weaknesses are that it exists only at the 80 bin centres, describes bin integrals rather than the density $d\sigma/dm_{hh}$, and does not use the fact that the coefficient functions are smooth. We propose to make it the primary model in a smooth form: each coefficient function is written as a sum of spline basis functions, the likelihood compares the *integral* of the model over each bin with POWHEG, a smoothness prior with strengths set by the evidence regularises the fit, and an optional threshold factor builds in the phase-space behaviour at $m_{hh} = 2m_h$. The model stays linear in its parameters, so the posterior, the predictions for any binning, the density at any m_{hh} and their uncertainties are all exact and cheap. Validation includes a new test of within-bin accuracy: fit on 30-GeV bins, predict the 15-GeV bins. The decisions to be made are collected in section 11.

The proposal in one paragraph

$\frac{d\sigma}{dm}(m;\theta) = \sum_{i=1}^{23} C_i(\theta)\, a_i(m)$, $a_i(m) = \phi(m) \sum_{j=1}^{J} \beta_{ij}\, B_j(m)$, with B_j cubic B-splines, ϕ an optional threshold factor, and $23J$ parameters β_{ij}. The cross section in bin $[e_k, e_{k+1}]$ is $\sum_{i,j} C_i(\theta)\, \beta_{ij}\, I_{jk}$ with $I_{jk} = \int_{e_k}^{e_{k+1}} \phi B_j\, dm$, linear in $\boldsymbol{\beta}$. Gaussian likelihood with the exact Poisson variances; Gaussian smoothness prior on the second differences of each $\beta_{i\cdot}$, its 23 strengths chosen by maximising the evidence; exact Gaussian posterior; the cancellation-aware discrepancy added to the predictions.

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1 Why this model

- **The network is not the limit.** The free 23×80 fit matches the ensemble on the test spectra (median χ^2/N 1.088 vs 1.085) and, with the cancellation-aware discrepancy, is equally well calibrated (pull s.d. 0.999 vs 1.004) (`docs/modelerror/new_error_model.pdf`). The remaining error is a mismatch between POWHEG and the 23-term form, which no model of that form can remove.
- **Linear is better, other things equal.** A model linear in its parameters has an exact Gaussian posterior: no Markov chains, no Laplace approximation, no random seeds, no ensemble. The uncertainty is a matrix inverse.
- **But the free fit has three defects.** (i) It is defined only at the bin centres; (ii) it models bin integrals, so reading it as $15\,\text{GeV} \times d\sigma/dm$ at the centre is an approximation (0.17% of the value, 6% of the noise, in the first ten bins); (iii) it treats the 80 values of each a_i as unrelated, so its a_i jitter from bin to bin and its poorly determined combinations (a_{16}–a_{20}) are not helped by their neighbours.
- **The network's interpolation is not to be trusted** between bin centres below 600 GeV: the 20 networks disagree there twice as much as at the centres, because nothing in the data constrains them.

The smooth model removes the three defects and keeps linearity. The free fit is its limit with one box-shaped basis function per bin and no prior (section 2.4).

2 The model

2.1 The density and the bins

POWHEG gives, for each spectrum s at HEFT point θ_s, the cross section in bins of 15 GeV, $t_{sk} = \int_{e_k}^{e_{k+1}} \frac{d\sigma}{dm}(m;\theta_s)\,dm$ plus noise. We model the density

$$\frac{d\sigma}{dm}(m;\theta) = \sum_{i=1}^{23} C_i(\theta)\, a_i(m), \tag{1}$$

with the same 23 monomials C_i as `HEFTNet` (`docs/dihiggs_model.pdf`); here $a_i(m)$ is a density, in pb/GeV. The model's prediction for bin k is the integral of eq. (1) over the bin, never its value at the centre.

2.2 The coefficient functions

Each coefficient function is expanded in J cubic B-splines on a fixed set of knots, times a threshold factor:

$$a_i(m) = \phi(m) \sum_{j=1}^{J} \beta_{ij}\, B_j(m). \tag{2}$$

B-splines in brief

A cubic B-spline $B_j(m)$ is a smooth bump, a piecewise cubic polynomial that is non-zero only over four consecutive knot intervals and joins with continuous first and second derivatives. A weighted sum of them can represent any smooth curve whose wiggles are not finer than the knot spacing. Each coefficient β_{ij} acts locally: it moves a_i only near knot j. They are the standard tool for smoothing in statistics (de Boor; Eilers & Marx).

Knots. The functions change fastest just above threshold and slowly above 600 GeV (`docs/paper/fig_coefficients.pdf`). The knots should therefore be denser near threshold. A simple choice is to space them uniformly in $u = \sqrt{m - 250\,\text{GeV}}$, which places a third of them below about 400 GeV. With a smoothness prior (section 4) the exact number matters little as long as it is generous: we propose $J = 40$ (920 parameters), and will check $J = 30$ and 60.

Threshold factor. Near $m_{hh} = 2m_h = 250$ GeV the phase space of two Higgs bosons vanishes like $\beta_h = \sqrt{1 - 4m_h^2/m^2}$, and so does $d\sigma/dm$ (for an s-wave). Choosing $\phi(m) = \beta_h(m)$ builds this in, so the splines need only describe the smooth remainder. The first 15-GeV bin (255–270 GeV) starts 5 GeV above threshold, so the factor matters mostly there; we will compare $\phi = \beta_h$ with $\phi = 1$ (section 11).

2.3 Linear in the parameters

Collect the $23J$ coefficients into one vector $\boldsymbol{\beta}$. The prediction for spectrum s in bin k is

$$f_{sk} = \sum_{i=1}^{23}\sum_{j=1}^{J} C_i(\theta_s)\, I_{jk}\, \beta_{ij} = \big(\boldsymbol{C}(\theta_s) \otimes \boldsymbol{I}_k\big)^{\mathsf{T}} \boldsymbol{\beta}, \qquad I_{jk} = \int_{e_k}^{e_{k+1}} \phi(m) B_j(m)\, dm, \tag{3}$$

where $\boldsymbol{I}_k = (I_{1k}, \ldots, I_{Jk})$ and $\otimes$ is the Kronecker product. The $80 \times J$ numbers I_{jk} depend only on the knots and the binning and are computed once, accurately, by Gauss–Legendre quadrature (8 points per bin per spline piece).

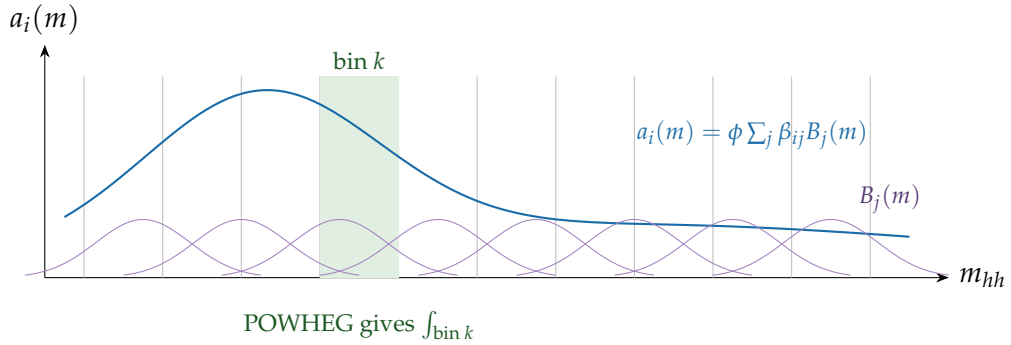


Figure 1: The coefficient function (blue) is a weighted sum of local spline bumps (purple). The data constrain its integral over each bin (green), not its value at the centre.

2.4 The free fit as a limit

If the basis functions are boxes, one per bin, $B_k(m) = 1$ inside bin k and 0 outside, and $\phi = 1$, then $I_{jk} = 15\,\text{GeV} \times \delta_{jk}$ and eq. (3) is the free 23×80 fit. The smooth model replaces the boxes by smooth bumps and adds a prior that prefers smooth a_i. The free fit is the special case, and will be kept as the benchmark.

3 The likelihood

The counts are Poisson (`docs/counts/counts_note.pdf`); with about 1,000 events in the bins near threshold and at least tens elsewhere, a Gaussian likelihood is accurate:

$$-\ln p(\boldsymbol{t} \mid \boldsymbol{\beta}) = \tfrac{1}{2} \sum_{s,k} w_{sk} \, (t_{sk} - f_{sk})^2 + \text{const}, \qquad w_{sk} = 1/\sigma_{sk}^2. \tag{4}$$

Weights, first pass. $\sigma_{sk}^2 = q_s(t_{sk} + q_s)$, the variance used so far (the extra q_s^2 protects nearly empty bins). Empty bins, restored by `counts.add_empty_bins`, enter with $t = 0$.

Weights, second pass (proposed). The model error is not small: for the most cancelling spectra it exceeds the noise. Fitting with the noise alone gives those bins too much weight. A second pass uses $\sigma_{sk}^2 = q_s(t_{sk} + q_s) + \varepsilon(m_k)^2 S_{sk}^2$, with $S_{sk} = \sum_i |C_i \bar{a}_{ik}|$ and ε from the first pass (section 6). This is generalised least squares with the error model we believe; it should mostly change the poorly determined combinations. Whether to iterate further is decided by whether ε changes.

Which rows. The training rows of the current split (86.4% of the rows), plus the restored empty bins; the 1.6% validation rows are not needed (there is no early stopping) and are added to the training rows; the 12% quick-test rows are kept for the discrepancy; the 1,000 test spectra for validation.

4 The smoothness prior and the evidence

4.1 The prior

For each coefficient function i, a Gaussian prior on the second differences of its spline coefficients (a "P-spline", Eilers & Marx 1996):

$$-\ln \pi(\boldsymbol{\beta}_i \mid \lambda_i) = \tfrac{1}{2}\lambda_i \sum_{j=3}^{J} (\beta_{ij} - 2\beta_{i,j-1} + \beta_{i,j-2})^2 + \tfrac{1}{2}\lambda_0 |\boldsymbol{\beta}_i|^2 + \text{const.} \tag{5}$$

The first term penalises curvature; it is the discrete analogue of $\int (a_i'')^2\, dm$ and is equivalent to an integrated random-walk Gaussian process. It leaves straight lines unpenalised, so a tiny ridge term λ_0 (fixed, 10^{-8} of the typical λ_i) makes the prior proper. One strength λ_i per function: the 23 functions differ by orders of magnitude in size and smoothness (`fig_coefficients.pdf`).

4.2 The posterior

Likelihood and prior are Gaussian in $\boldsymbol{\beta}$, so the posterior is exactly Gaussian, with

$$H = \Phi^{\mathsf{T}} W \Phi + \Lambda, \qquad \hat{\boldsymbol{\beta}} = H^{-1} \Phi^{\mathsf{T}} W \boldsymbol{t}, \tag{6}$$

where Φ has one row $(\boldsymbol{C}_s \otimes \boldsymbol{I}_k)^{\mathsf{T}}$ per data row and $\Lambda = \text{blockdiag}(\lambda_i D^{\mathsf{T}} D + \lambda_0 I)$. The structure of eq. (3) makes $\Phi^{\mathsf{T}} W \Phi$ cheap, as for the Laplace calculation:

$$\Phi^{\mathsf{T}} W \Phi = \sum_{k=1}^{80} S_k \otimes \boldsymbol{I}_k \boldsymbol{I}_k^{\mathsf{T}}, \qquad S_k = \sum_s w_{sk}\, \boldsymbol{C}_s \boldsymbol{C}_s^{\mathsf{T}}\ (23 \times 23), \tag{7}$$

80 Kronecker products of a 23×23 and a $J \times J$ matrix. With $J = 40$, H is 920×920; everything takes seconds on a laptop.

4.3 Choosing the smoothness by the evidence

The marginal likelihood of the data given λ is available in closed form:

$$\ln p(\boldsymbol{t} \mid \lambda) = -\tfrac{1}{2}\chi^2(\hat{\boldsymbol{\beta}}) - \tfrac{1}{2}\hat{\boldsymbol{\beta}}^{\mathsf{T}} \Lambda \hat{\boldsymbol{\beta}} + \tfrac{1}{2} \ln\det \Lambda - \tfrac{1}{2} \ln\det H + \text{const}, \tag{8}$$

($\ln\det\Lambda$ over the proper part). We maximise it over the 23 λ_i with MacKay's fixed-point updates, already written for `laplace.py`: $\lambda_i \leftarrow \gamma_i / (\hat{\boldsymbol{\beta}}_i^{\mathsf{T}} D^{\mathsf{T}} D \hat{\boldsymbol{\beta}}_i)$, with γ_i the number of well-determined combinations in function i. Unlike the network case, the model is exactly linear, so the evidence is exact, not an approximation.

Evidence and model error

The evidence assumes the likelihood is right. With noise-only weights, the model error makes the data look rougher than the true a_i, and the evidence may choose too little smoothing. The second-pass weights (section 3) address this. As a check, we will also choose the λ_i by 10-fold cross-validation over spectra and compare.

5 Predictions and uncertainties

For any θ and any interval $[m_1, m_2]$, the cross section and its variance are

$$\sigma_{[m_1,m_2]} = \boldsymbol{g}^{\mathsf{T}}\hat{\boldsymbol{\beta}}, \qquad \mathrm{Var} = \boldsymbol{g}^{\mathsf{T}}H^{-1}\boldsymbol{g}, \qquad \boldsymbol{g} = \boldsymbol{C}(\theta) \otimes \int_{m_1}^{m_2} \phi \boldsymbol{B}\, dm, \tag{9}$$

and likewise for the density at any m (replace the integral by $\phi(m)\boldsymbol{B}(m)$), for the coefficient functions $a_i(m)$ with their bands, and for the total cross section. Correlations between bins, absent from the bin-by-bin network uncertainty, come out automatically. The predictive uncertainty of a new POWHEG spectrum adds the Poisson noise and the discrepancy (section 6).

6 The discrepancy

As established in `new_error_model.pdf`: $s^2_{\mathrm{disc}} = \varepsilon(m)^2(\sum_i |C_i a_i|)^2$, with $\varepsilon(m)$ fitted on the quick-test rows by requiring $\langle z^2 \rangle = 1$ bin by bin, then smoothed. For predictions over wider intervals (e.g. 30 GeV, or the total cross section) the discrepancy of adjacent bins must not be added as if independent: the residuals are correlated over 5–8 bins near threshold. For this plan we treat the discrepancy as fully correlated within 60 GeV and independent beyond, as a conservative interim rule, and flag a Gaussian-process discrepancy as the follow-up.

7 Validation

1. **The standard tests** on the 1,000 test spectra, in the format used so far: median χ^2/N, χ^2 per bin below and above 555 GeV, pull s.d. overall, by region and by cancellation fifth, $|z| < 2$; against the free fit and the network ensemble.
2. **Within-bin accuracy: the merged-bin test.** Merge adjacent pairs of bins in the training data (30-GeV bins), fit, and predict the original 15-GeV bins of the test spectra. The free fit can only split each 30-GeV value in half; the smooth model predicts the split from the shape. The metric is χ^2 and pulls of the 15-GeV predictions, especially below 600 GeV. This is the direct test of $d\sigma/dm$ inside a bin.
3. **Edge-bin test.** The POWHEG source files have 100 bins of 15 GeV from 0 GeV (`data/README.md`). The bins 240–255 and 1455–1500 GeV are there but unused so far; predicting them from a fit without them tests the threshold factor and the behaviour at the high-mass end.
4. **Smoothness and conditioning.** Roughness of the a_i (rms second difference), the number of well-determined combinations γ_i per function, and whether the poorly determined a_{16}–a_{20} improve.
5. **Sensitivity**: J = 30, 40, 60; $\phi = 1$ vs β_h; evidence vs cross-validation; first- vs second-pass weights.
6. **Residual diagnostics**: correlation of residuals between bins, as in `model_error_note.pdf`; the smooth model should not reduce it (it is a property of the data), and must not increase it.

Success criteria

Test-spectrum accuracy no worse than the free fit (median $\chi^2/N \leq 1.09$); calibrated pulls with the discrepancy (s.d. within 2% of 1 overall and in every cancellation fifth); in the merged-bin

test, a clear improvement over the halved free fit below 600 GeV; and smooth a_i with bands, defined at every m_{hh}.

8 Code

- **New module** `dihiggs/smooth.py`, commented for non-experts:

```
basis = SplineBasis(knots, threshold=True)        # B_j(m) * phi(m)
I = basis.bin_integrals(edges)                    # [J, n_bins], by quadrature
model = SmoothHEFT(basis, n_coeffs=23)
model.fit(C, bins, t, var, n_iter=20)             # exact posterior; lambda by evidence
f, v = model.predict(C_new, edges_new)            # any binning, mean and variance
d, vd = model.density(C_new, m_grid)              # d sigma / d m at any m
a, va = model.coefficients(m_grid)                # the 23 a_i(m) with variances
lnZ = model.log_evidence()
```

  The 23 coefficients $C_i(\theta)$ come from `laplace.wilson_coefficients`, so the model uses exactly the monomials of `HEFTNet`.

- **New notebook** `code/heft_smooth.ipynb`: fit (both passes), discrepancy, all validation tests, figures in the paper format; results in `runs/SMOOTH_<tag>/` (the posterior mean and covariance, about 7 MB for $J = 40$, and the summaries).

- **A test with an exact answer**: a toy with three monomials and known smooth a_i, generated with Poisson noise, which the fit must recover within its quoted uncertainties (pull s.d. 1).

- **Export**: the model is a $23J$-vector and the knot list; a small C++/Python function evaluates it, simpler than exporting a network (`heft_export.ipynb`).

9 What changes in the paper

The primary model becomes the smooth 23×80 model, with its exact posterior and the cancellation-aware discrepancy. The network ensemble and the free fit become cross-checks; the Bayesian-network study becomes a short methods remark (why HMC under-sampled, how the Laplace mixture agrees). The appendix is rewritten around the new model, the figures are remade (spectra panel, coefficient functions now as continuous curves with bands, all test spectra), and the executive summary and roadmap are updated.

10 Order of work

1. `smooth.py` and the toy test (CPU, here).
2. First pass on the real data with $J = 40$, $\phi = \beta_h$; standard tests; comparison with the free fit. *Gate:* accuracy no worse than the free fit.
3. Discrepancy, second pass, calibration by cancellation fifth.
4. Merged-bin and edge-bin tests; sensitivity studies.
5. Write-up (`docs/smooth/smooth_results.pdf`); then the paper changes.

All of it runs on a laptop CPU in minutes; nothing needs the 5090.

11 Decisions for you

Decisions before I proceed

1. **Cubic B-splines with $J = 40$ knots spaced uniformly in $\sqrt{m - 250}$**, with J = 30 and 60 as checks. Agreed?
2. **Threshold factor $\phi = \beta_h$** as the default, $\phi = 1$ as a check. Agreed?
3. **P-spline prior** (second differences), one strength per function, chosen by the evidence with cross-validation as a check. Or would you prefer an explicit Gaussian process with a fitted correlation length?
4. **Two-pass weights**, the second including the discrepancy. Agreed?
5. **Fit range**: the 80 bins of 255–1455 GeV as now, or include the bins 240–255 and 1455–1500 GeV from the source files (which would require extending `heft_prepare_data`)? I propose: keep 255–1455 for the fit and use the edge bins only as an extrapolation test.
6. **Interim correlation rule** for the discrepancy over wide intervals (fully correlated within 60 GeV), with a Gaussian-process discrepancy as the follow-up. Agreed?
7. **New module, notebook and folders**: `dihiggs/smooth.py`, `code/heft_smooth.ipynb`, `runs/SMOOTH_*`, `docs/smooth/`. Acceptable?